

University of Stuttgart
Germany
Institute of Industrial Automation and
Software Engineering

Probabilistic Action Prediction of Human Drivers Using Dynamic Bayesian Networks

Master Thesis 1111

**Submitted at the University of Stuttgart by
Amal Chulliyat Jose**

Electrical Engineering

Examiner: Prof. Dr.-Ing. Andrey Morozov

Supervisor: Georgios Katranis, M.Sc.

15 April 2026



Contents

List of Figures	iii
List of Tables	iv
List of Abbreviations	v
Glossary	vii
Kurzfassung	ix
Abstract	xi
1. Introduction	1
2. Theoretical Background	3
2.1. Probability Theory	3
2.1.1. Bayes' Theorem	4
2.2. Probabilistic Graphical Models	5
2.2.1. Bayesian Networks	5
2.2.2. Dynamic Bayesian Networks	6
2.3. Inference in Chain-Structured DBNs	7
2.3.1. Forward-Backward Algorithm	7
2.4. Parameter Learning using Maximum Likelihood Estimation	9
2.4.1. Expectation-Maximization Algorithm	10
3. State of the Art	12
3.1. Rule-Based Models	12
3.1.1. Car-Following Models	12
3.1.2. Lane-Change Models	13
4. Conclusions	14
Bibliography	15
A. Example appendix chapter	17

List of Figures

2.1. Example of a Bayesian Network (BN)	6
2.2. Two-slice representation of a Dynamic Bayesian Network (DBN).	6

List of Tables

List of Abbreviations

AV Autonomous Vehicle

HDV Human-Driven Vehicle

DBN Dynamic Bayesian Network

BN Bayesian Network

DAG Directed Acyclic Graph

MAP Maximum A Posteriori

AI Artificial Intelligence

THW Time Headway

TTC Time to Collision

KI Künstliche Intelligenz

CPU Central Processing Unit

SAV Semi-Autonomous Vehicle

V2X Vehicle-to-Everything

List of Abbreviations

RV Random Variable

CPD Conditional Probability Distribution

CPT Conditional Probability Tables

HMM Hidden Markov Model

EM Expectation-Maximization

MLE Maximum Likelihood Estimation

ADAS Advanced Driver Assistance Systems

IDM Intelligent Driver Model

FVDM Full Velocity Difference Model

OVM Optimal Velocity Model

MOBIL Minimizing Overall Braking Induced by Lane Changes

Glossary

Autonomous Vehicle (AV) A vehicle that can drive by itself without human control. It uses sensors and artificial intelligence to understand the surrounding environment and make driving decisions.

Belief State The probability distribution over the latent states given the observed data at a specific time step.

Confusion Matrix A matrix showing the performance of a classification model, across the true and predicted classes.

Directed Acyclic Graph (DAG) A graph made up of nodes and directed edges between them, where the edges do not form any closed loops.

DBN A probabilistic graphical model that uses a directed acyclic graph to represent time-series data, where each node represents a random variable and the edges show how these variables depend on each other within and across time steps.

Emission Model The probability distribution that describes how observations are generated from latent states.

Entropy A measure of uncertainty in a probability distribution. It also quantifies the amount of information needed to describe the distribution.

F1 Score The harmonic mean of precision and recall.

Human-Driven Vehicle (HDV) A vehicle that is controlled by a human driver who performs all driving tasks such as steering, accelerating, and braking.

Inference The process of determining the most likely latent states from the observed data.

Latent State A variable that is not directly observed but is inferred from data.

Likelihood The probability of the observed data under a given model.

Maximum A Posteriori (MAP) The estimate of the most likely latent state, given the observed data. It is the state that maximizes the posterior distribution.

Mixed Traffic A traffic situation where both autonomous vehicles and human-driven vehicles operate together.

Glossary

Observation The data that is directly measured from the system, represented as feature vectors.

Posterior The probability distribution over the latent states after observing the data.

Precision The ratio of true positive predictions to the total predicted positives. It indicates the accuracy of positive predictions.

Prior The probability distribution over the latent states before observing any data.

Recall The ratio of true positive predictions to the total actual positives. It indicates the completeness of positive predictions.

Random Variable (RV) A variable that can take on different values, each with an associated probability.

State Transition Probability of transitioning from one latent state to another.

Time Headway (THW) The time needed for a vehicle to reach the current position of the vehicle ahead if it maintains its current velocity.

Time to Collision (TTC) The time needed for a vehicle to collide with the vehicle ahead if both of them maintain their current velocity.

Trajectory The time-ordered sequence describing the movement of a vehicle over time.

Kurzfassung

Künstliche Intelligenz (KI) verändert autonome Systeme durch bedeutende Fortschritte in den Bereichen Sensorik, Kommunikation und Rechenleistung. Obwohl vollständig autonome Fahrzeuge (Autonomous Vehicles (AVs)) voraussichtlich immer häufiger werden, werden sie auf absehbare Zeit gemeinsam mit von Menschen gesteuerten Fahrzeugen (Human-Driven Vehicles (HDVs)) am Straßenverkehr teilnehmen. Um in diesem gemischten Verkehrsumfeld sicher zu operieren, müssen autonome Systeme in der Lage sein, menschliche Fahrhandlungen zu modellieren und vorherzusagen.

Diese Arbeit adressiert dieses wichtige vorgelagerte Problem durch die Entwicklung eines transparenten, probabilistischen Rahmens zur Vorhersage hochrangiger menschlicher Fahrhandlungen unter Verwendung dynamischer Bayes'scher Netze (Dynamic Bayesian Networks (DBNs)). Ziel dieser Forschung ist die Entwicklung eines Modells, das zukünftige Fahreraktionen abschätzen kann und gleichzeitig Unsicherheiten in einer interpretierbaren Form darstellt. Ein solches Modell kann innerhalb eines AV-Systems als Belief-Modell dienen und Informationen für nachgelagerte Planungs- und sicherheitsrelevante Module bereitstellen.

Die Studie verwendet den naturalistischen Autobahnfahrdatensatz highD, um zeitliche Muster menschlichen Fahrverhaltens zu erlernen. Fahrzeugtrajektorien werden in sequenzielle, fensterbasierte Beobachtungen umgewandelt, die relevante Bewegungs- und Interaktionsinformationen über die Zeit erfassen. Diese Beobachtungen fassen zusammen, wie sich ein Fahrzeug und der umgebende Verkehr über kurze Zeitintervalle entwickeln. Anschließend wird ein DBN trainiert, um die zeitliche Entwicklung des Fahrerverhaltens zu modellieren und Aktionswahrscheinlichkeiten aus eingehenden Beobachtungen abzuleiten. Das vorgeschlagene DBN verwendet eine hierarchische latente Struktur, in der Aktionen von Fahrregimen abhängig sind. Genauer gesagt repräsentiert das Modell zwei latente Regime, ein Regime mit geringer Interaktion und ein Regime mit hoher Interaktion, und sagt Aktionszustände innerhalb jedes Regimes vorher. Dadurch kann das Modell sowohl das vorhergesagte Fahrmanöver als auch den Interaktionskontext darstellen, in dem es auftritt.

Eine zentrale Stärke dieses Ansatzes liegt in seiner transparenten probabilistischen White-Box-Struktur. Im Gegensatz zu vielen Black-Box-Lernmethoden erzeugt das DBN seine Vorhersagen über explizite probabilistische Beziehungen, die mithilfe der Wahrscheinlichkeitstheorie interpretiert werden können. Dies ist insbesondere für sicherheitsrelevante Anwendungen von Bedeutung, bei denen Interpretierbarkeit und die Darstellung von Unsicherheiten wichtig sind. Das Modell unterstützt daher nicht nur die Vorhersageleistung, sondern auch die Erklärbarkeit, was bei der Betrachtung solcher

Systeme für sicherheitskritische Verkehrsanwendungen von Vorteil ist. Die Studie leistet einen Beitrag zur Forschung im Bereich des autonomen Fahrens, indem sie Aktionsvorhersage, zeitliche probabilistische Schlussfolgerung und Interpretierbarkeit in einem einheitlichen Rahmen kombiniert.

Die Ergebnisse zeigen, dass das Modell in der Lage ist, aussagekräftige Verhaltensmuster aus naturalistischen Verkehrsdaten zu lernen und vielversprechende Manöर्वorhersagen über einen zukünftigen Zeithorizont zu generieren. In der aktuellen Evaluation erreichte das Modell eine exakte Next-Step-Genauigkeit von 46,4 %, eine Makropräzision von 44,3 % sowie eine Trefferquote von 68,0 % über einen Vorhersagehorizont von zwei Sekunden. Eine Vorhersage zum Zeitpunkt t wird dabei als Treffer gezählt, wenn das vorhergesagte Manöver innerhalb der nächsten fünf Fensterschritte auftritt, was etwa zwei Sekunden entspricht. Die durchschnittliche Time-to-Hit betrug 0,50 s und der Median 0,40 s, was zeigt, dass viele korrekte Vorhersagen früh innerhalb des Vorhersagehorizonts auftreten. Diese Ergebnisse deuten darauf hin, dass das Modell zukünftige menschliche Fahreraktionen innerhalb eines praktisch nutzbaren kurzfristigen Zeitintervalls antizipieren kann, anstatt nur einen einzelnen exakten zukünftigen Zeitpunkt vorherzusagen. Implementierungs-Benchmarks zeigen zudem geringe Online-Inferenzkosten mit einer mittleren CPU-Latenz von 0,164 ms pro Fahrzeug für Belief-Update und nächste Aktionsvorhersage.

Insgesamt zeigen die Ergebnisse, dass DBNs ein vielversprechender Ansatz für die probabilistische Vorhersage menschlicher Fahreraktionen sind. Die Studie demonstriert, dass hochrangige menschliche Fahrmanöver mithilfe naturalistischer Verkehrsdaten in einer interpretierbaren und zeitlich strukturierten Weise modelliert werden können. Dies stellt einen wichtigen Schritt hin zu einem besseren Verständnis menschlichen Fahrverhaltens im gemischten Verkehr dar und kann AV-Systeme dabei unterstützen, das Verhalten umliegender menschlicher Fahrer besser einzuschätzen.

Stichwörter: Vorhersage menschlichen Fahrverhaltens, Autonomes Fahren, Von Menschen gesteuerte Fahrzeuge, Dynamische Bayes'sche Netze, Manöर्वorhersage, Probabilistische Modellierung, Gemischte Verkehrsumgebungen

Abstract

Artificial Intelligence (AI) is transforming autonomous systems through major advances in sensing, communication, and computation. Although fully Autonomous Vehicles (AVs) are expected to become more common, they will share the road with Human-Driven Vehicles (HDVs) for the foreseeable future. To operate safely in this mixed-traffic setting, autonomous systems must be able to model and predict human driver actions.

This work addresses this important upstream problem by developing a transparent, probabilistic framework for predicting high-level human driving actions using Dynamic Bayesian Networks (DBNs). The purpose of this research is to develop a model that can estimate future driver actions while also representing uncertainty in an interpretable form. Such a model can act as a belief model inside an AV framework and provide information for downstream planning and safety-related modules.

The study uses the highD naturalistic highway driving dataset to learn temporal patterns of human driving behavior. Vehicle trajectory data are converted into sequential window-based observations that capture relevant motion and interaction information over time. These observations summarize how the vehicle and its surrounding traffic evolve over short time intervals. A DBN is then trained to model the temporal evolution of driver behavior and to infer action probabilities from incoming observations. The proposed DBN uses a hierarchical latent structure in which actions are conditioned on driving regimes. More specifically, the model represents two latent regimes, low-interaction and high-interaction, and predicts action states within each regime. This allows the framework to represent both the predicted maneuver and the interaction context in which it occurs.

A key strength of this approach is that it provides a transparent, white-box probabilistic structure. Unlike many black-box learning methods, the DBN makes predictions through explicit probabilistic relationships that can be interpreted using probability theory. This is valuable in safety-relevant applications, where interpretability and uncertainty representation are important. The model therefore supports not only prediction, but also explainability, which is useful when such systems are considered for use in safety-critical transport settings. The study contributes to science and engineering by combining action prediction, temporal probabilistic reasoning, and interpretability in a single framework for autonomous driving applications.

The results show that the model can learn meaningful behavioral patterns from naturalistic traffic data and generate promising maneuver predictions over a future horizon. In the current evaluation, the model achieved 46.4% exact next-step accuracy, 44.3% macro precision, and a 68.0% hit rate over a 2-second prediction horizon. Here, a prediction

Abstract

made at time t is counted as a hit if the predicted maneuver appears at any point within the next 5 window steps, corresponding to about 2 seconds. The mean time-to-hit was 0.50 s, and the median time-to-hit was 0.40 s, showing that many correct predictions occur early within the horizon. These results suggest that the model can anticipate future human driver actions within a practically useful short-term interval rather than only at one exact future instant. Implementation benchmarks also indicate low online inference cost, with a mean single-vehicle Central Processing Unit (CPU) latency of 0.164 ms for belief update and next-action prediction.

Overall, the findings indicate that DBNs are a promising framework for probabilistic human driver action prediction. The study shows that high-level human driving maneuvers can be modeled in an interpretable and temporally structured way using naturalistic traffic data. This provides a useful step toward better behavior understanding in mixed traffic and may support AV systems that need to reason about nearby human drivers.

Keywords: Human Driver Behavior Prediction, Autonomous Driving, Human-Driven Vehicles, Dynamic Bayesian Networks, Maneuver Prediction, Probabilistic Modeling, Mixed Traffic Environments

1. Introduction

Automation is influencing almost every part of daily life, and transportation is one of the areas changing most rapidly. Autonomous Vehicles (AVs) are at the forefront of this change, revolutionizing the transportation sector. Their development has accelerated because of recent progress in Artificial Intelligence (AI), sensing systems, computing, and communication technologies. These innovations could enhance traffic efficiency, increase travel comfort, improve road safety, and, most importantly, reshape future mobility.

However, fully autonomous traffic systems do not yet exist anywhere in the world. AVs will have to operate in mixed traffic together with Semi-Autonomous Vehicles (SAVs) and Human-Driven Vehicles (HDVs). This creates important challenges because different road users provide the different level of information. AVs typically rely on rich on-board sensing and potentially have Vehicle-to-Everything (V2X) communication, while HDVs do not explicitly communicate their future intentions and often behave in a stochastic way [1][2]. As a result, the internal goals, decisions, and reactions of human drivers cannot be observed directly. They must instead be inferred from observable signals such as vehicle motion, lane changes, relative positions, and interactions with nearby vehicles.

For this reason, human driver behavior prediction is an important problem in autonomous driving. If an AV can estimate what nearby human drivers are likely to do next, it can make safer and better-informed decisions. For example, knowledge of likely braking, lane-changing, or car-following behavior can support downstream tasks such as risk assessment, trajectory prediction, and motion planning [3].

Over the last two decades, the researchers have developed three broad families of behavioral models for road users: rule-based models, probabilistic models, and learning-based methods [4] [5] [6]. In parallel, several game-theoretic decision-making paradigms for AVs have emerged [7]. Rule-based models are simple and easy to understand, but they are often too rigid to capture the variability of real human behavior. Learning-based methods can learn complex patterns from large datasets, but many of them are difficult to interpret and often behave like black boxes. Probabilistic models provide a useful middle ground because they can explicitly represent uncertainty, capture temporal dependencies, and still remain interpretable.

This thesis focuses on probabilistic modeling of high-level human driving behavior in highway traffic. Rather than building the full control or planning stack of an AV, this work addresses an important upstream problem: predicting the likely future maneuvers of surrounding human drivers. The goal is to develop a model that can estimate future driver actions while also representing uncertainty in a clear and interpretable way.

1. Introduction

To achieve this, the thesis uses a Dynamic Bayesian Network (DBN) to model the temporal relationship between observable driving features and hidden behavioral states. Since driver intentions are not directly observable, the model represents them through latent variables and infers them probabilistically from trajectory-based observations [8]. In particular, the DBN is designed to capture latent factors such as driving context and action dynamics, and to produce probabilistic predictions of future high-level maneuvers. The model is trained and evaluated using the highD dataset, a naturalistic highway driving dataset containing detailed vehicle trajectory recordings [9]. From these trajectories, relevant features are extracted to describe motion, surrounding traffic context, and interaction behavior. These features serve as observations for the DBN.

The thesis studies whether such a probabilistic framework can provide accurate, interpretable, and computationally efficient short-horizon predictions of human driving actions. In particular, it examines the suitability of the dataset, the structure of the DBN, the usefulness of the selected observation features, the quality of future action prediction, the interpretability of the learned states, and the runtime performance of the model for online use.

2. Theoretical Background

2.1. Probability Theory

Human driving behaviour is inherently stochastic. In a traffic scenario, different drivers may interpret the same situation differently depending on their perception, experience, and intentions. As a result, multiple maneuvers may be possible under the same traffic conditions. For example, a driver approaching a slower vehicle may choose to brake, maintain speed, or initiate a lane change. Since such decisions cannot be determined deterministically, behaviour prediction must account for uncertainty. Probability theory provides a mathematical framework for representing and reasoning about uncertainty. Instead of predicting a single deterministic outcome, probabilistic models represent a distribution over possible outcomes and their likelihoods. This allows the model to capture the variability and uncertainty inherent in human decision-making.

In probability theory, each uncertain quantity is represented as a Random Variable (RV) [10]. In the context of driver behaviour modelling, variables such as vehicle velocity, acceleration, or lane position can be treated as observable RVs. In addition, latent quantities such as driver style or driving maneuver intention can also be represented as RVs that must be inferred from observations.

Complex systems require modelling interactions between many RVs. When several RVs are considered together, their behaviour can be described using a joint probability distribution [10]. For two RVs, O and Z , representing the observable RV and latent RV, the joint probability is written as $P(O, Z)$ and represents the probability that both variables take specific values at the same time. This idea can be extended to any number of variables. In driver behaviour modelling, joint probability is important because observable variables and latent variables influence each other and must often be described together within a unified probabilistic framework.

It is also important to describe dependencies between RVs. This is done using conditional probability [10]. The conditional probability of O given Z is defined as

$$P(O | Z) = \frac{P(O, Z)}{P(Z)},$$

provided that $P(Z) > 0$. Conditional probability describes how the probability of one RV changes when information about another RV is known. This concept is essential because the probability of a future driving maneuver is estimated conditioned on the observed traffic context during inference. During model learning, the probability of observations conditioned on latent states is also required.

2. Theoretical Background

Another important concept is marginal probability, which gives the probability of a subset of RVs irrespective of the values of the others [10]. For example, if the joint distribution $P(O, Z)$ is known, the marginal probability of Z can be obtained by summing over all possible values of O :

$$P(Z) = \sum_O P(O, Z).$$

In addition to describing dependencies between RVs, it is also useful to identify situations where variables do not influence each other. Two RVs are said to be independent if the value of one variable does not affect the probability of the other. Formally, two variables A and B are independent if their joint probability can be written as

$$P(A, B) = P(A)P(B).$$

In many practical problems, variables are not completely independent but may become independent when conditioned on another variable. This property is known as conditional independence. Two variables A and B are conditionally independent given a third variable C if the joint conditional distribution factorizes as

$$P(A, B | C) = P(A | C)P(B | C).$$

Conditional independence is a key concept in probabilistic graphical models because it allows complex joint probability distributions to be represented using simpler conditional relationships between variables [11].

2.1.1. Bayes' Theorem

A fundamental principle in probabilistic reasoning is Bayes' theorem, which allows beliefs about uncertain events to be updated when new observations become available [10]. It relates conditional and marginal probabilities and is expressed as

$$P(Z | O) = \frac{P(O | Z)P(Z)}{P(O)} \quad (2.1)$$

Here, $P(Z)$ is called the prior probability. The prior represents the initial belief about a variable before observing any new data. In the context of driver behaviour modelling, the prior can represent the initial probability of different latent driver states before considering the current traffic situation. The term $P(O | Z)$ is called the likelihood. The likelihood describes how probable the observed data O is, given a particular latent state Z . In driver behaviour prediction, it measures how compatible the observed vehicle motion and traffic features are with a given latent driver state. The term $P(O)$ is called the evidence (or marginal likelihood). It represents the total probability of observing the data O under all possible values of Z . The evidence acts as a normalization factor that ensures the resulting probabilities sum to one. Finally, $P(Z | O)$ is called the posterior probability. The posterior represents the updated belief about Z after observing O . In

2. Theoretical Background

behaviour modelling, the posterior corresponds to the updated probability distribution over the latent driver state given the observed traffic context.

Bayes' theorem therefore provides a systematic way to update probabilistic beliefs when new information becomes available. This principle forms the basis for probabilistic inference in many models used for behaviour prediction, including DBN.

2.2. Probabilistic Graphical Models

Probabilistic graphical models provide an efficient way to represent complex joint probability distributions over many RVs [10] [11]. As the number of variables increases, modelling the full joint distribution becomes difficult because the dependencies between variables also increase. Probabilistic graphical models address this problem by representing RVs and their probabilistic relationships using a graph. In this representation, nodes correspond to RVs, while edges describe dependencies between them. The graph structure also encodes conditional independence assumptions. These assumptions allow the joint probability distribution to be factorised into simpler Conditional Probability Distributions (CPDs).

There are two main types of probabilistic graphical models:

1. **Directed graphical models**, where edges have a direction and represent conditional dependencies between variables.
2. **Undirected graphical models**, where edges do not have a direction and represent symmetric relationships between variables.

One important class of probabilistic graphical models is the Bayesian Network (BN). Its temporal extension is the DBN, which is particularly suitable for modelling sequential processes over time.

2.2.1. Bayesian Networks

A BN is a directed probabilistic graphical model that represents a set of RVs and their conditional dependencies using a Directed Acyclic Graph (DAG) [11]. Each variable in the network is associated with a CPD, which describes how the variable depends on its parent variables. These CPDs are often represented using Conditional Probability Tables (CPTs). If a variable has no parents in the graph, its distribution is given by a prior probability distribution.

In general, the joint probability distribution of all variables in a BN can be factorised according to the graph structure. For a set of variables $X_1, X_2, \dots, X_n$, the joint distribution can be written as

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)) \quad (2.2)$$

2. Theoretical Background

Here, $\text{Pa}(X_i)$ denotes the set of parent variables of X_i in the graph. This factorisation reflects the conditional independence assumption embedded in the network structure.

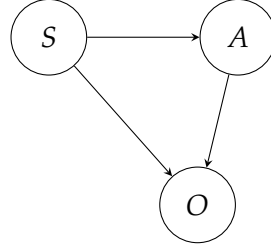


Figure 2.1.: Example of a BN

For the example network shown in Figure 2.1, the joint distribution can be written as

$$P(S, A, O) = P(S)P(A | S)P(O | S, A).$$

Standard BNs describe relationships between variables within a single time step. However, in many real-world problems, such as driver behaviour modelling, the system evolves over time. In such cases, a temporal extension of BNs, called a DBN, is used.

2.2.2. Dynamic Bayesian Networks

In a DBN, the system is represented as a sequence of time-indexed RVs that describe how the system evolves over time. A common representation of a DBN uses a two-slice temporal model that captures dependencies both within a single time step and across consecutive time steps. This representation typically relies on the first-order Markov assumption, which states that the current state depends only on the previous state and not on earlier states.

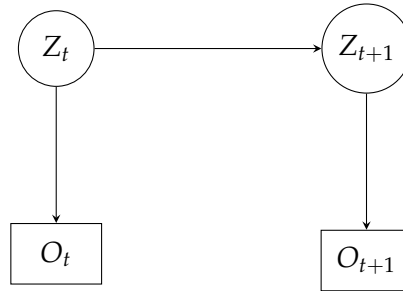


Figure 2.2.: Two-slice representation of a DBN.

In the example shown in Figure 2.2, let Z_t denote the latent state at time step t and O_t denote the corresponding observation. Under the Markov assumption, the joint probability distribution over a sequence of states and observations can be written as

2. Theoretical Background

$$P(Z_{1:T}, O_{1:T}) = P(Z_1) \prod_{t=2}^T P(Z_t | Z_{t-1}) \prod_{t=1}^T P(O_t | Z_t). \quad (2.3)$$

In this formulation, $P(Z_t | Z_{t-1})$ represents the state transition model, which captures how the latent state evolves over time, while $P(O_t | Z_t)$ represents the emission model that relates the hidden state to the observed data.

In the context of driver behaviour modelling, DBNs provide a probabilistic framework for representing latent driver states and their temporal evolution based on observed vehicle motion and traffic context.

2.3. Inference in Chain-Structured DBNs

Inference in probabilistic graphical models refers to computing the probability distribution of unknown variables given observed evidence. In temporal models such as DBNs, this involves estimating the distribution of latent states from a sequence of observations [10] [11].

For the DBN shown in Figure 2.2, the latent variable Z_t may represent hidden quantities such as driver style or maneuver state. These variables are not directly observable and must therefore be inferred from measurable quantities such as vehicle motion and the surrounding traffic context. Inference is required because the model contains hidden states, whereas only the observation sequence $O_{1:T}$ is available.

In temporal probabilistic models, three common inference tasks arise [12]:

1. **Filtering:** Estimating the current latent state using observations up to the current time step, $P(Z_t | O_{1:t})$.
2. **Smoothing:** Estimating the latent state using the full observation sequence, $P(Z_t | O_{1:T})$.
3. **Prediction:** Estimating future latent states from the observations available up to the current time, $P(Z_{t+k} | O_{1:t})$.

For the chain-structured DBN shown in Figure 2.2, the latent state at time t depends only on the previous state Z_{t-1} , and the observation O_t depends only on the current state Z_t . This structure corresponds to a first-order Markov model with conditionally independent observations, which is equivalent to the structure of a Hidden Markov Model (HMM). Therefore, exact inference can be performed using the forward-backward algorithm [13]. This algorithm is widely used in HMMs, which are a special case of DBNs.

2.3.1. Forward–Backward Algorithm

The Forward–Backward algorithm computes posterior probabilities of latent states by combining information from past and future observations [13]. More specifically, it esti-

2. Theoretical Background

mates the posterior distribution of the hidden state at each time step given the complete observation sequence.

Let the observation sequence be $O_{1:T} = \{O_1, O_2, \dots, O_T\}$ and the corresponding latent states be $Z_{1:T}$. The main objective is to compute the posterior distribution, $P(Z_t | O_{1:T})$, which is called the smoothing distribution.

The Forward–Backward algorithm performs this computation using two recursive quantities: the forward messages and the backward messages.

Forward Pass

The forward variable $\alpha_t(z_t)$ represents the joint probability of observing the partial sequence up to time t and being in state z_t :

$$\alpha_t(z_t) = P(O_{1:t}, Z_t = z_t) \quad (2.4)$$

It can be computed recursively as

$$\alpha_t(z_t) = P(O_t | Z_t = z_t) \sum_{z_{t-1}} P(Z_t = z_t | Z_{t-1} = z_{t-1}) \alpha_{t-1}(z_{t-1}) \quad (2.5)$$

with initialization

$$\alpha_1(z_1) = P(Z_1 = z_1) P(O_1 | Z_1 = z_1)$$

Here, $P(Z_t = z_t | Z_{t-1} = z_{t-1})$ is the transition probability, $P(O_t | Z_t = z_t)$ is the observation likelihood, and $P(Z_1 = z_1)$ is the initial state distribution.

Backward Pass

The backward variable $\beta_t(z_t)$ represents the probability of observing the remaining sequence from time $t + 1$ to T given the current state:

$$\beta_t(z_t) = P(O_{t+1:T} | Z_t = z_t) \quad (2.6)$$

It is computed recursively in the reverse direction as

$$\beta_t(z_t) = \sum_{z_{t+1}} P(Z_{t+1} = z_{t+1} | Z_t = z_t) P(O_{t+1} | Z_{t+1} = z_{t+1}) \beta_{t+1}(z_{t+1}) \quad (2.7)$$

with initialization

$$\beta_T(z_T) = 1$$

Posterior State Estimation

Once the forward and backward messages have been computed, the posterior probability of the hidden state at time step t can be calculated. This quantity is commonly denoted

2. Theoretical Background

as $\gamma_t(z_t)$ and represents the probability that the system is in state z_t at time t given the full observation sequence:

$$\gamma_t(z_t) = P(Z_t = z_t \mid O_{1:T})$$

Using the forward and backward variables, this can be written as

$$\gamma_t(z_t) = \frac{\alpha_t(z_t)\beta_t(z_t)}{P(O_{1:T})} \quad (2.8)$$

where $P(O_{1:T})$ is the likelihood of the full observation sequence. Using the forward and backward variables, it can be written as

$$P(O_{1:T}) = \sum_{z_t} \alpha_t(z_t)\beta_t(z_t).$$

for any time step t .

The Forward–Backward algorithm therefore combines information from earlier observations through the forward pass and from later observations through the backward pass.

Posterior Transition Probability

In addition to the posterior probability of individual states, it is often useful to compute the joint posterior probability of two consecutive states. This quantity gives the posterior probability of a transition from state z_t to state z_{t+1} given the full observation sequence:

$$\xi_t(z_t, z_{t+1}) = P(Z_t = z_t, Z_{t+1} = z_{t+1} \mid O_{1:T})$$

Using the forward and backward variables, this can be written as

$$\xi_t(z_t, z_{t+1}) = \frac{\alpha_t(z_t) P(Z_{t+1} = z_{t+1} \mid Z_t = z_t) P(O_{t+1} \mid Z_{t+1} = z_{t+1}) \beta_{t+1}(z_{t+1})}{P(O_{1:T})} \quad (2.9)$$

This posterior transition probability is useful in parameter learning, especially in Expectation-Maximization (EM) based training methods [13].

2.4. Parameter Learning using Maximum Likelihood Estimation

A probabilistic model contains several unknown parameters that must be learned from data. For a chain-structured temporal model such as the one shown in Figure 2.2, these parameters can be collectively written as

$$\theta = \{\pi, A, B\},$$

2. Theoretical Background

where π denotes the initial state distribution, A is the state transition matrix, and B represents the parameters of the emission model.

If both the observation sequence $O_{1:T}$ and the latent state sequence $Z_{1:T}$ were known, parameter learning could be performed by maximizing the complete-data likelihood

$$\theta^* = \arg \max_{\theta} P(O_{1:T}, Z_{1:T} \mid \theta) \quad (2.10)$$

However, in latent-variable models the state sequence $Z_{1:T}$ is not observed. Therefore, learning must be based only on the observed data, which gives the Maximum Likelihood Estimation (MLE) objective

$$\theta^* = \arg \max_{\theta} P(O_{1:T} \mid \theta) \quad (2.11)$$

Since the latent state sequence is unknown, the likelihood of the observations is obtained by marginalizing over all possible latent state sequences:

$$P(O_{1:T} \mid \theta) = \sum_{Z_{1:T}} P(O_{1:T}, Z_{1:T} \mid \theta) \quad (2.12)$$

As shown in (2.12), the observed-data likelihood is obtained by summing the joint probability over all possible latent state sequences. Because this summation involves many possible latent state sequences, direct maximization of the observed-data likelihood in (2.11) is difficult. To address this problem, the EM algorithm is commonly used [12]. For chain-structured models such as HMMs, this learning procedure is known as the Baum–Welch algorithm [13].

2.4.1. Expectation–Maximization Algorithm

The EM algorithm is an iterative method for parameter estimation in probabilistic models with latent variables. It addresses the difficulty of maximizing the observed-data likelihood by alternating between estimating the latent states and updating the model parameters.

Expectation Step

In the expectation step (E-step), the current parameter estimate is used to compute the posterior distribution of the latent states given the observations. For the chain-structured model considered here, this is done using the Forward–Backward algorithm described by (2.5) and (2.7). The E-step provides the posterior state probabilities

$$\gamma_t(i) = P(Z_t = i \mid O_{1:T})$$

and the posterior transition probabilities

$$\xi_t(i, j) = P(Z_t = i, Z_{t+1} = j \mid O_{1:T}).$$

As shown in (2.8) and (2.9), these posterior quantities represent the probabilities of state occupancy and state transitions conditioned on the full observation sequence.

2. Theoretical Background

Maximization Step

In the maximization step (M-step), the model parameters are updated using the posterior quantities computed in the E-step.

For discrete-state chain-structured models, the initial state distribution is updated as

$$\pi_i = \gamma_1(i) \quad (2.13)$$

The transition probabilities can be re-estimated as

$$A_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \quad (2.14)$$

The parameters of the emission model are updated using the observations weighted by the posterior state probabilities. The exact form of this update depends on the chosen emission model.

The E-step and M-step are repeated until convergence. Under standard EM, each iteration does not decrease the observed-data likelihood, and the algorithm converges to a local optimum or stationary point [12].

3. State of the Art

Existing approaches for modelling and predicting driver behaviour can be broadly grouped into rule-based models, probabilistic models, and learning-based data-driven models. In addition, hybrid methods combine elements from more than one of these families in order to exploit their complementary strengths.

3.1. Rule-Based Models

Rule-based models characterize driving behaviour through explicit mathematical or logical rules. These models relate observable variables such as spacing, relative velocity, and desired speed to longitudinal acceleration or lane-change decisions. Their main advantages are interpretability and low computational cost [14]. For this reason, they are widely used in traffic simulation, Advanced Driver Assistance Systems (ADAS), and baseline behaviour modelling. However, as they rely on hand-crafted assumptions about driver decision-making, they often represent behaviour in a deterministic or simplified manner.

3.1.1. Car-Following Models

Intelligent Driver Model

The Intelligent Driver Model (IDM) is one of the most widely used car-following models [14]. It expresses a vehicle's acceleration as a continuous function of its current velocity, desired velocity, spacing to the leading vehicle, and relative velocity [15]. The formulation also includes parameters related to desired time headway and comfortable braking, which make the model suitable for safety-oriented traffic simulation. Because of its simple structure and physically meaningful parameters, the IDM remains a common reference model in intelligent driving research.

Gipps' Model

Gipps' model is one of the earliest rule-based car-following models. It is based on the idea that a driver chooses a speed that is both desirable and safe with respect to the preceding vehicle [16]. This safety-oriented formulation allows the model to capture conservative driving behaviour and ensures collision-free motion under typical traffic conditions.

Optimal Velocity Model and Full Velocity Difference Model

Another classical approach is the Optimal Velocity Model (OVM), in which the driver's acceleration depends on the difference between the current velocity and an optimal velocity determined by the spacing to the leading vehicle [17]. This model captures the driver's intuition to accelerate when the gap ahead increases and decelerate when it decreases. An extension of the OVM is the Full Velocity Difference Model (FVDM), where, instead of only considering the gap between the vehicles, the model also captures the velocity difference between the ego and preceding vehicles when calculating the acceleration [18]. This helps reduce the oscillations observed in the original OVM.

3.1.2. Lane-Change Models

MOBIL Model

The Minimizing Overall Braking Induced by Lane Changes (MOBIL) model is a lane-changing model built upon the IDM to describe lateral driving behaviour [14] [19]. It incorporates incentive and safety criteria to evaluate the feasibility of lane changes. The incentive criterion indicates whether a lane change could be advantageous in terms of the driver's acceleration, whereas the safety criterion ensures that the lane change does not cause unnecessary deceleration for surrounding vehicles.

Despite their interpretability and efficiency, rule-based models have limited ability to represent uncertainty and inter-driver variability in complex traffic scenarios. These limitations motivate the use of probabilistic and learning-based approaches.

4. Conclusions

Bibliography

- [1] W. Schwarting, J. Alonso-Mora, and D. Rus, "Planning and decision-making for autonomous vehicles", *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 1, no. 1, pp. 187–210, 2018.
- [2] M. R. Abedi, Z. Rashidi, N. Mokari, H. Saeedi, and N. Zorba, *Precise hdv positioning through safety-aware integrated sensing and communication in a value-of-information-driven 6g v2x system*, 2025. arXiv: 2510.02363 [eess.SY].
- [3] N. A. Madjid, A. Ahmad, M. Mebrahtu, Y. Babaa, A. Nasser, S. Malik, B. Hassan, N. Werghi, J. Dias, and M. Khonji, "Trajectory prediction for autonomous driving: Progress, limitations, and future directions", *Information Fusion*, vol. 126, p. 103 588, Feb. 2026.
- [4] N. M. Negash and J. Yang, "Driver behavior modeling toward autonomous vehicles: Comprehensive review", *IEEE Access*, vol. 11, pp. 22 788–22 821, 2023.
- [5] A. Rudenko, L. Palmieri, M. Herman, K. M. Kitani, D. M. Gavrila, and K. O. Arras, "Human motion trajectory prediction: A survey", *The International Journal of Robotics Research*, vol. 39, no. 8, pp. 895–935, Jun. 2020.
- [6] S. Lefevre, D. Vasquez, and C. Laugier, "A survey on motion prediction and risk assessment for intelligent vehicles", *Robomech Journal*, vol. 1, Jul. 2014.
- [7] Y. Yang, N. Negash, and J. Yang, "Recent advances in interactive driving of autonomous vehicles: Comprehensive review of approaches", *Automotive Innovation*, vol. 8, Apr. 2025.
- [8] K. Murphy, *Dynamic Bayesian Networks: Representation, Inference and Learning*. University of California, Berkeley, 2002.
- [9] R. Krajewski, J. Bock, L. Kloecker, and L. Eckstein, *The highd dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems*, 2018. arXiv: 1810.05642 [cs.CV].
- [10] K. P. Murphy, *Probabilistic Machine Learning: An Introduction*. MIT Press, 2022.
- [11] D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques* (Adaptive computation and machine learning). MIT Press, 2009.
- [12] K. P. Murphy, "Machine learning - a probabilistic perspective", in *Adaptive computation and machine learning series*, 2012.
- [13] L. Rabiner, "A tutorial on hidden markov models and selected applications in speech recognition", *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

Bibliography

- [14] S. Zhou, S. Zheng, J. Tian, R. Jiang, and H. M. Zhang, *Twenty-five years of the intelligent driver model: Foundations, extensions, applications, and future directions*, 2025. arXiv: 2506.05909 [physics.soc-ph].
- [15] M. Treiber, A. Hennecke, and D. Helbing, “Congested traffic states in empirical observations and microscopic simulations”, *Phys. Rev. E*, vol. 62, pp. 1805–1824, 2 Aug. 2000.
- [16] P. Gipps, “A behavioural car-following model for computer simulation”, *Transportation Research Part B: Methodological*, vol. 15, no. 2, pp. 105–111, 1981.
- [17] M. Bando, K. Hasebe, A. Nakayama, A. Shibata, and Y. Sugiyama, “Dynamical model of traffic congestion and numerical simulation”, *Phys. Rev. E*, vol. 51, pp. 1035–1042, 2 Feb. 1995.
- [18] R. Jiang, Q. Wu, and Z. Zhu, “Full velocity difference model for a car-following theory”, *Phys. Rev. E*, vol. 64, p. 017 101, 1 Jun. 2001.
- [19] A. Kesting, M. Treiber, and D. Helbing, “General lane-changing model mobil for car-following models”, *Transportation Research Record*, vol. 1999, pp. 86–94, 2007.

A. Example appendix chapter

Declaration of Compliance

I hereby declare to have written this work independently and to have respected in its preparation the relevant provisions, in particular those corresponding to the copyright protection of external materials. Whenever external materials (such as images, drawings, text passages) are used in this work, I declare that these materials are referenced accordingly (e. g. quote, source) and, whenever necessary, consent from the author to use such materials in my work has been obtained.

Signature:

Stuttgart, on the 15 April 2026